INCH-POUND

MIL-M-28787/529A
23 March 1994
SUPERSEDING
MIL-M-28787/529
24 October 1991

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

* MIL-STD-1397 TYPE D SERIAL INTERFACE ADAPTER MODULE, KEY CODE AJL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

* 1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic MIL-STD-1397 type D serial interface adapter module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

* Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Code 603A, Bldg. 2917, NAVSURFWARCENDIV, 300 Highway 361, Crane, IN 47522-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.142 failures per million operating hours for class I or 2.045 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be AJL.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Pin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:37Z
Check the source to verify that this is the currentversion before use.

TABLE I. Module electrical requirements.

	Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
					Min	Max	Min	Max	Min	Max	
	A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
*		End-of-life	II	S	1	INF	1	INF	1	INF	MOhm
	A2-A51	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
*		End-of-life	II	X	1	INF	1	INF	1	INF	MOhm
*	B1-B7	Continuity	III	X	0	1	0	1	0	1	Ohm
	C	Power supply stress	IV	S	----------Module shall pass all remaining tests----------						
	D1	Ibd	V	S	-0.022	1.02	-0.022	1.02	-0.022	1.02	mA
*	D2-D8	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
*	D7	Ibd	V	S	-0.582	1.14	-0.582	1.14	-0.582	1.14	mA
*	D8	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
*	D9,D10	Ibd	V	S	-0.022	1.02	-0.022	1.02	-0.022	1.02	mA
*	D11	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
*	D12	Ibd	V	S	-0.022	1.02	-0.022	1.02	-0.022	1.02	mA
*	D13	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
	E1-E16	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
	F1	Iih	VII	X	-3.04	-0.503	-3.04	-0.503	-3.04	-0.503	mA
	F2-F6	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
*	F7	Iih	VII	X	-6.64	-0.353	-6.64	-0.353	-6.64	-0.353	mA
	F8	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
	F9,F10	Iih	VII	X	-3.04	-0.503	-3.04	-0.503	-3.04	-0.503	mA
	F11	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
	F12	Iih	VII	X	-3.04	-0.503	-3.04	-0.503	-3.04	-0.503	mA
	F13	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
	G1	Iil	VIII	X	-3.04	-0.503	-3.04	-0.503	-3.04	-0.503	mA
	G2-G6	Iil	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
*	G7	Iil	VIII	X	-6.64	-0.353	-6.64	-0.353	-6.64	-0.353	mA
	G8	Iil	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
	G9,G10	Iil	VIII	X	-3.04	-0.503	-3.04	-0.503	-3.04	-0.503	mA
	G11	Iil	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
	G12	Iil	VIII	X	-3.04	-0.503	-3.04	-0.503	-3.04	-0.503	mA
	G13	Iil	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA

* See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T18:37Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
H1-H16	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J16	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
L1-L8	Vic	XI	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
* M1	PSFC	XII	S	1.9	3.6	1.9	3.6	1.9	3.6	µF
* M2	PSFC	XII	S	3.8	7.2	3.8	7.2	3.8	7.2	µF
* M3	PSFC	XII	S	14.4	26.7	14.4	26.7	14.4	26.7	µF
* N1	Icc	XIII	X	450	1150	450	1150	450	1150	mA
N2	Icc	XIII	X	-263	-130	-263	-130	-263	-130	mA
P1,P2	Functional	XIV	X	See table XIV						
Q1,Q2	Dynamic	XV	S	See table XV						
Q3,Q4	Dynamic	XV	Q	See table XV						
R1-R6	Cin	XVII	Q	2	75	2	75	2	75	pF
R7	Cin	XVII	Q	8	200	8	200	8	200	pF
R8	Cin	XVII	Q	2	100	2	100	2	100	pF
R9	Cin	XVII	Q	4	75	4	75	4	75	pF
R10	Cin	XVII	Q	4	100	4	100	4	100	pF
R11	Cin	XVII	Q	2	75	2	75	2	75	pF
R12	Cin	XVII	Q	4	100	4	100	4	100	pF
R13	Cin	XVII	Q	2	125	2	125	2	125	pF
T1,T2	PSTA	XVIII	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T18:37Z
Check the source to verify that this is the current version before use.

1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:37Z
Check the source to verify that this is the current version before use.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/529-1 for class I and M28787/529-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc and (V2) of -5.7 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

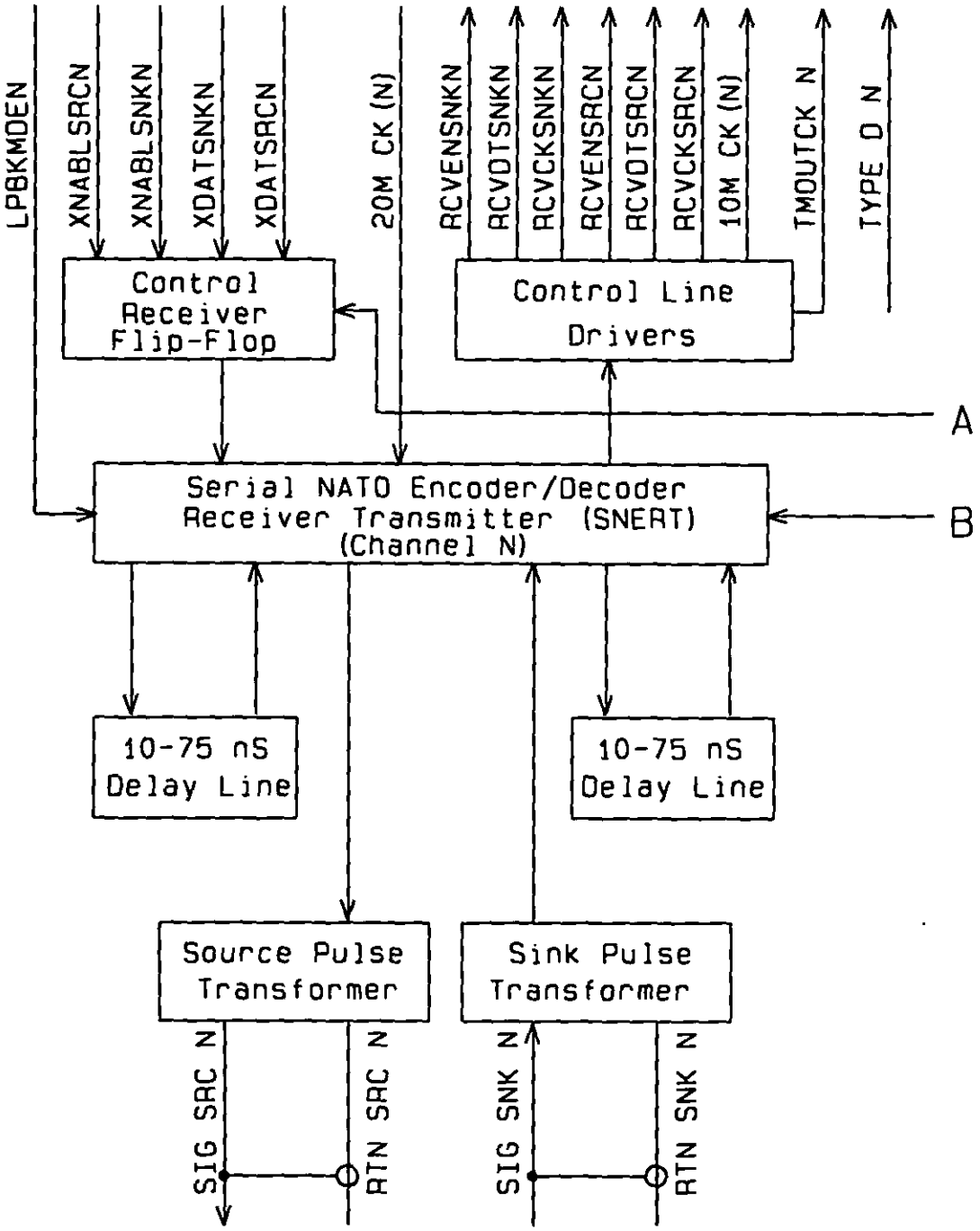
4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

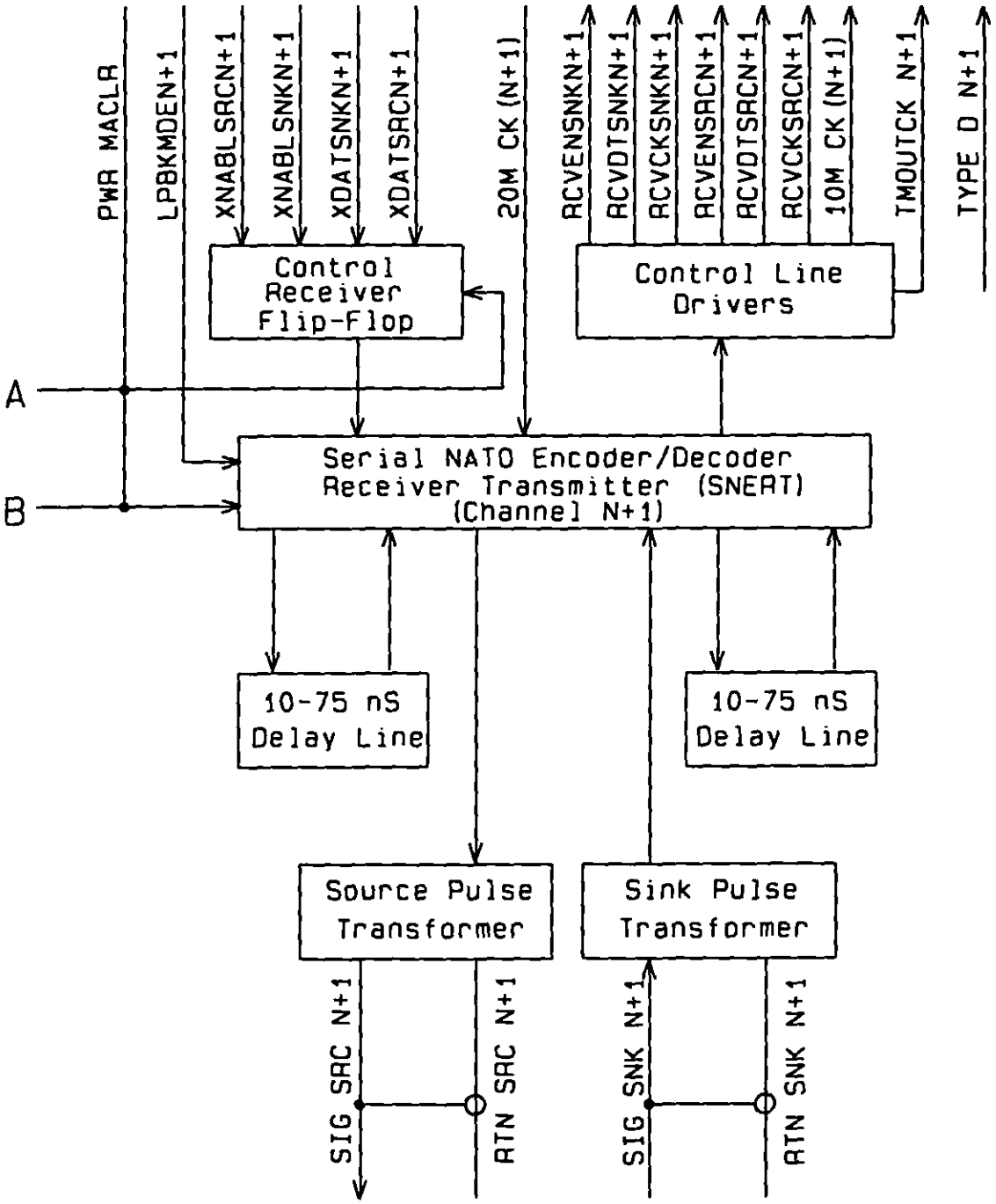
4.4 Test procedures. The module shall be tested as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:37Z
Check the source to verify that this is the current version before use.



* **FIGURE 1. AJL functional block diagram.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.



* FIGURE 1. AJL functional block diagram - Continued.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T18:37Z
Check the source to verify that this is the current version before use.

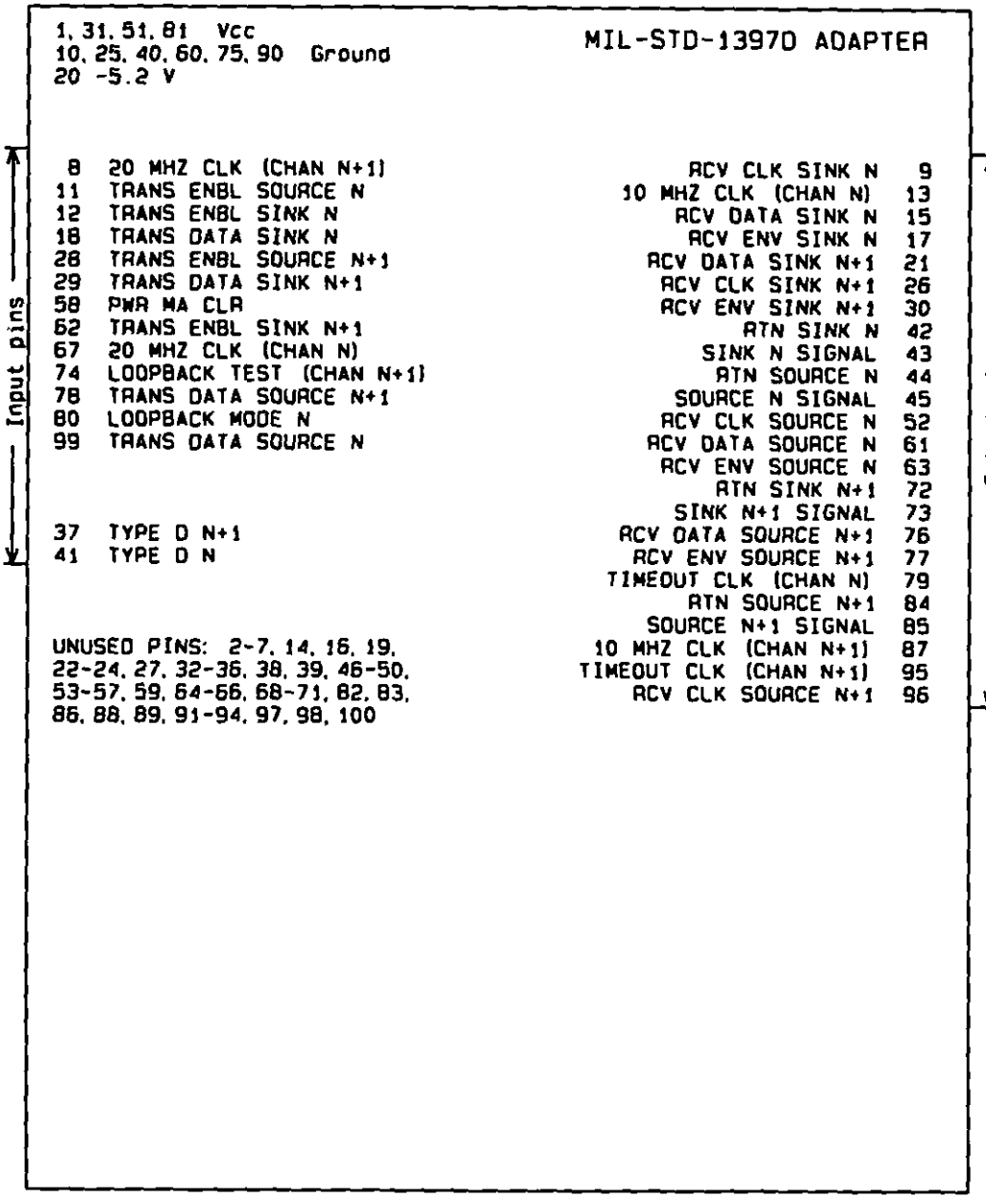


FIGURE 2. AJL pin assignment.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:37Z
Check the source to verify that this is the current version before use.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

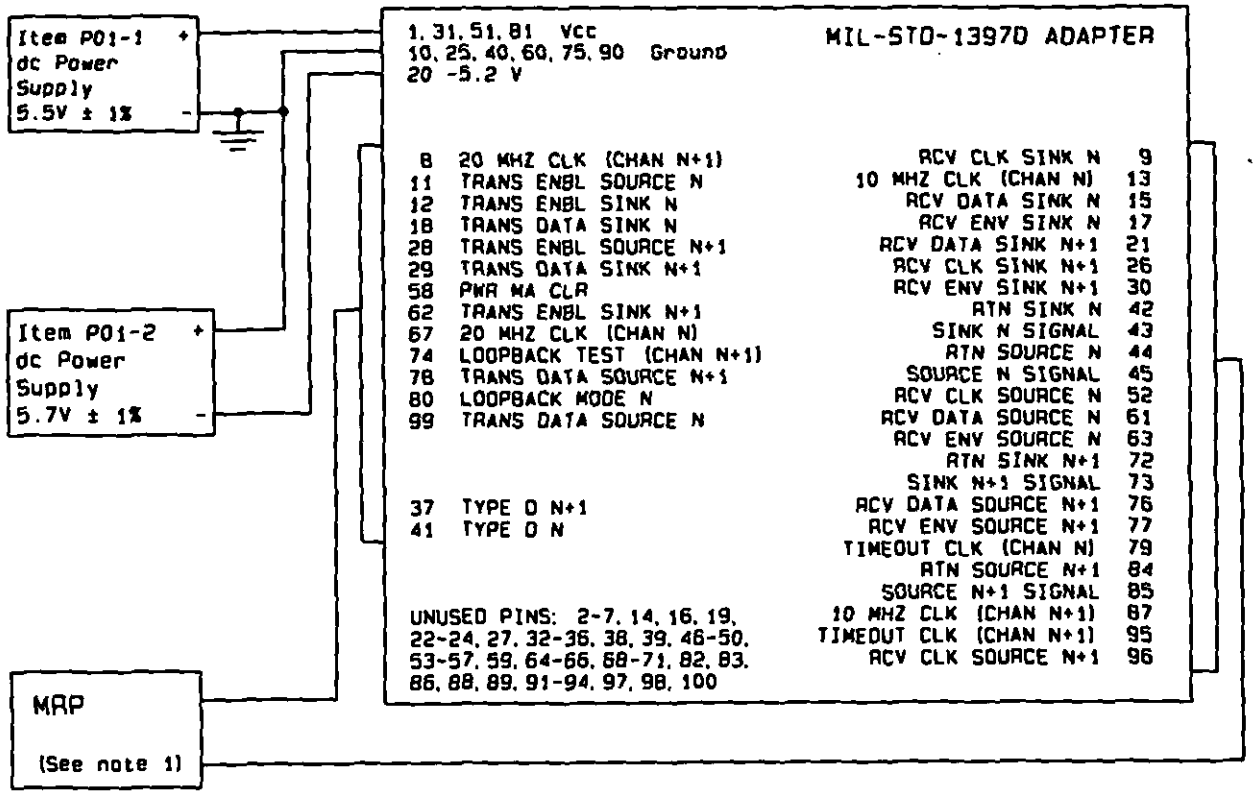
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:37Z
Check the source to verify that this is the current version before use.



NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. AJL operating temperature and component temperature test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

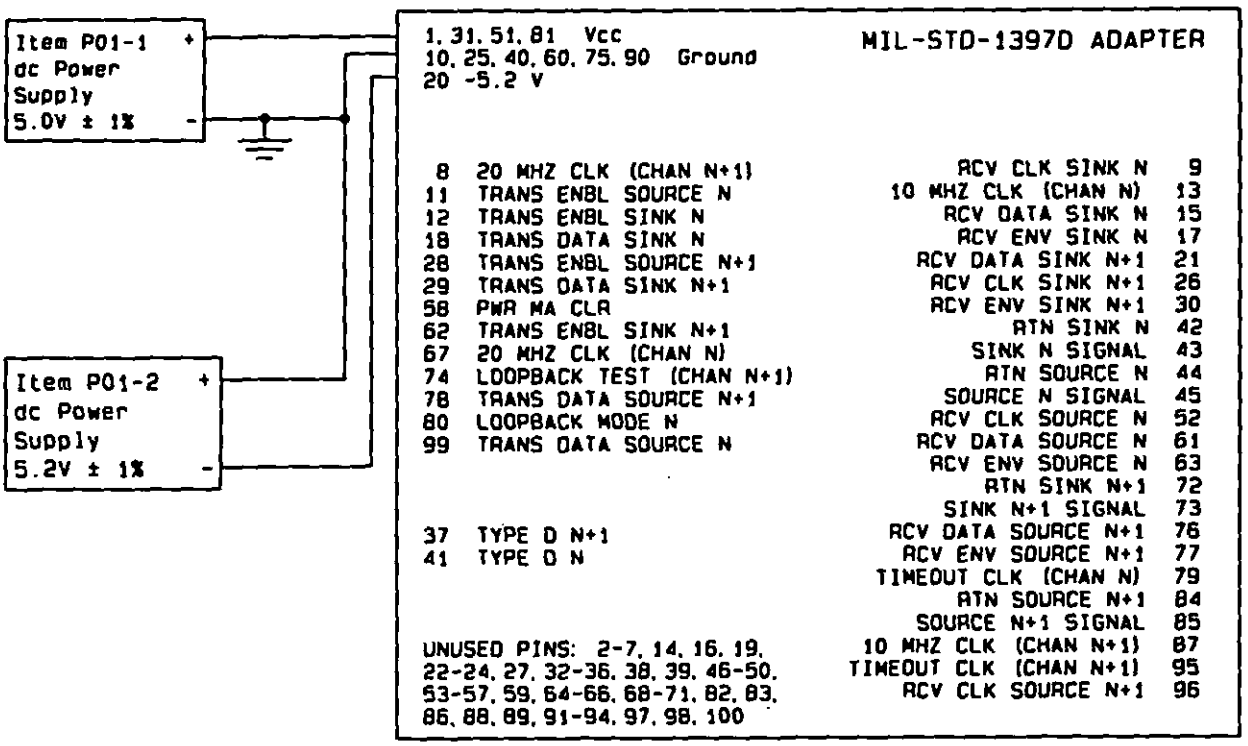


FIGURE 4. AJL vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T18:37Z
Check the source to verify that this is the current version before use.

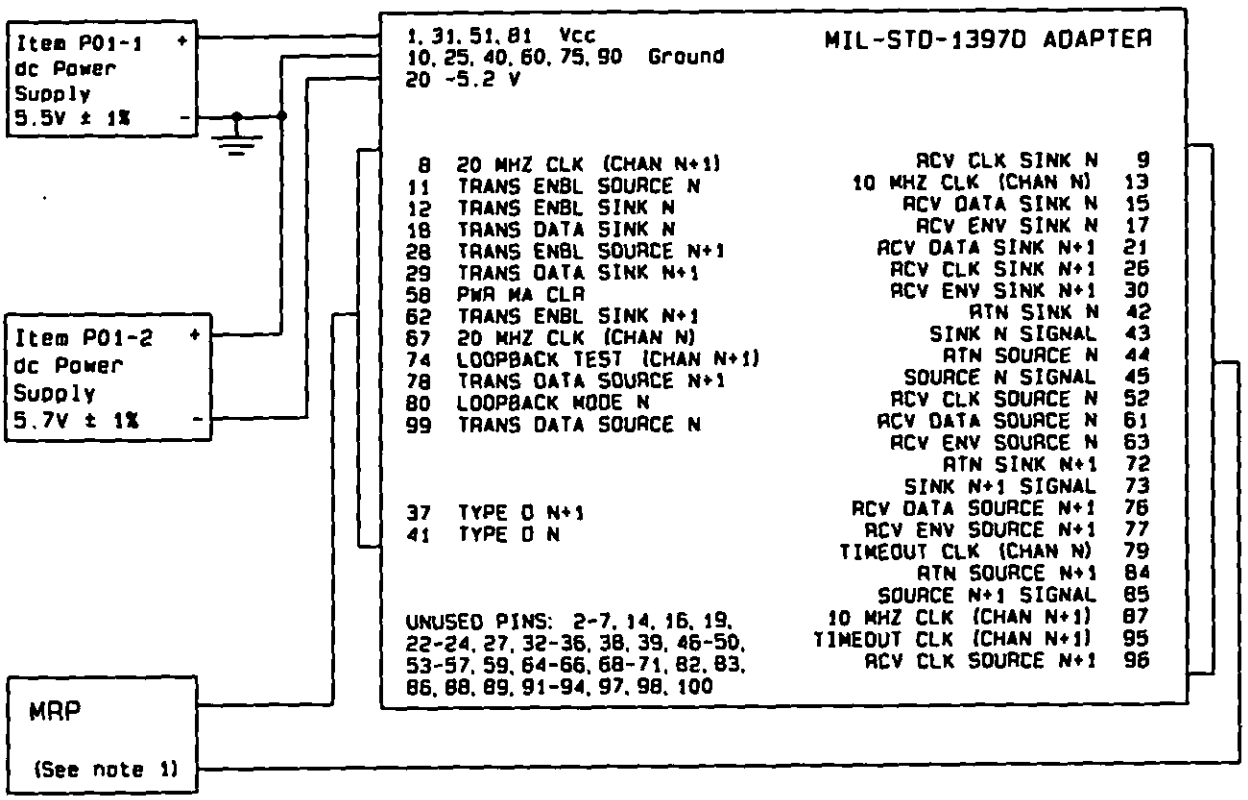
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.



NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. AJL life test circuit.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:37Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ibd = input breakdown current.
(c) Icc = power supply current.
(d) Iih = input high level current.
(e) Iil = input low level current.
(f) Iin = input forcing current.
(g) Ioh = output high level current.
(h) Iol = output low level current.
(i) Ios = output short circuit current.
(j) NA = not applicable.
(k) PSFC = power supply filter capacitance.
(l) PSTA = power supply transient amplitude.
(m) Vic = input clamp diode voltage.
(n) Vih = input high level voltage.
(o) Vil = input low level voltage.
(p) Vin = input forcing voltage.
(q) Voh = output high level voltage.
(r) Vol = output low level voltage.
(s) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: (Vcc) 4.5 V dc to 5.5 V dc and (V2) -4.7 V dc to -5.7 V dc. Module power dissipation: 7.33 W (max); 4.98 W (typical).

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:37Z
Check the source to verify that this is the current version before use.

6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
20M CK(N+1)	NA	0.4	2.4	-3.04	-0.503	1.02	75
XNABLSRCN	-1.5	0.4	2.4	-0.4	0.02	0.1	75
XNABLSNKN	-1.5	0.4	2.4	-0.4	0.02	0.1	75
XDATSNKN	-1.5	0.4	2.4	-0.4	0.02	0.1	75
XNABLSRCN+1	-1.5	0.4	2.4	-0.4	0.02	0.1	75
XDATSNKN+1	-1.5	0.4	2.4	-0.4	0.02	0.1	75
* PWR MACLR	NA	0.4	2.4	-6.64	-0.353	1.14	200
XNABLSNKN+1	-1.5	0.4	2.4	-0.4	0.02	0.1	100
20M CK(N)	NA	0.4	2.4	-3.04	-0.503	1.02	75
LPBKMDEN+1	NA	0.4	2.4	-3.04	-0.503	1.02	100
XDATSRCN+1	-1.5	0.4	2.4	-0.4	0.02	0.1	75
LPBKMDEN	NA	0.4	2.4	-3.04	-0.503	1.02	100
XDATSRCN	-1.5	0.4	2.4	-0.4	0.02	0.1	125

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:37Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)
RCVCKSNKN	2.4	0.5	-0.4	6.0	-40.0
10M CK(N)	2.4	0.5	-0.4	6.0	-40.0
RCVDTSNKN	2.4	0.5	-0.4	6.0	-40.0
RCVENSNKN	2.4	0.5	-0.4	6.0	-40.0
RCVDTSNKN+1	2.4	0.5	-0.4	6.0	-40.0
RCVCKSNKN+1	2.4	0.5	-0.4	6.0	-40.0
RCVENSNKN+1	2.4	0.5	-0.4	6.0	-40.0
RCVCKSRCN	2.4	0.5	-0.4	6.0	-40.0
RCVDTSRCN	2.4	0.5	-0.4	6.0	-40.0
RCVENSRCN	2.4	0.5	-0.4	6.0	-40.0
RCVDTSRCN+1	2.4	0.5	-0.4	6.0	-40.0
RCVENSRCN+1	2.4	0.5	-0.4	6.0	-40.0
TMOUTCK N	2.4	0.5	-0.4	6.0	-40.0

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:37Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)
10M CK(N+1)	2.4	0.5	-0.4	6.0	-40.0
TMOUTCK N+1	2.4	0.5	-0.4	6.0	-40.0
RCVCKSRCN+1	2.4	0.5	-0.4	6.0	-40.0
RTN SNK N	2.4	0.5	NA	NA	NA
SIG SNK N	2.4	0.5	NA	NA	NA
RTN SRC N	2.4	0.5	NA	NA	NA
SIG SRC N	2.4	0.5	NA	NA	NA
RTN SRC N+1	NA	NA	NA	NA	NA
SIG SRC N+1	NA	NA	NA	NA	NA
RTN SNK N+1	NA	NA	NA	NA	NA
SIG SNK N+1	NA	NA	NA	NA	NA

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	RTN SRC N to RCVENSRCN	-	-	-	56	56	56	nS	12
		SIG SRC N to RCVENSRCN	-	-	-					
		RTN SRC N+1 to RCVENSRCN+1	-	-	-					
1		SIG SRC N+1 to RCVENSRCN+1	-	-	-	56	56	56		
2		RTN SRC N to RCVENSRCN	-	-	-	206	206	206		
		SIG SRC N to RCVENSRCN	-	-	-					
		RTN SRC N+1 to RCVENSRCN+1	-	-	-					
2		SIG SRC N+1 to RCVENSRCN+1	-	-	-	206	206	206		
3		RTN SRC N to RCVCKSRCN	-	-	-	176	176	176		
		SIG SRC N to RCVCKSRCN	-	-	-					
3	tpd	RTN SRC N+1 to RCVCKSRCN+1	-	-	-	176	176	176	nS	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	SIG SRC N+1 to RCVCKSRCN+1	-	-	-	176	176	176	nS	12
4		RTN SNK N to RCVENSNKN	-	-	-	56	56	56		
		SIG SNK N to RCVENSNKN	-	-	-					
		RTN SNK N+1 to RCVENSNKN+1	-	-	-					
4		SIG SNK N+1 to RCVENSNKN+1	-	-	-	56	56	56		
5		RTN SNK N to RCVENSNKN	-	-	-	206	206	206		
		SIG SNK N to RCVENSNKN	-	-	-					
		RTN SNK N+1 to RCVENSNKN+1	-	-	-					
5		SIG SNK N+1 to RCVENSNKN+1	-	-	-	206	206	206		
6	tpd	RTN SNK N to RCVCKSNKN	-	-	-	176	176	176	nS	12

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:37Z
Check the source to verify that this is the current version before use.

6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	SIG SNK N to RCVCKSNKN	-	-	-	176	176	176	nS	12
		RTN SNK N+1 to RCVCKSNKN+1	-	-	-					
6		SIG SNK N+1 to RCVCKSNKN+1	-	-	-	176	176	176		
7		20M CK(N) to 10M CK(N)	-	-	-	50	50	50		
7		20M CK(N+1) to 10M CK(N+1)	-	-	-	50	50	50		
8		20M CK(N) to 10M CK(N)	-	-	-	59	59	59		
8		20M CK(N+1) to 10M CK(N+1)	-	-	-	59	59	59		
9		20M CK(N) to TMOUTCK N	-	-	-	60	60	60		
9		20M CK(N+1) to TMOUTCK N+1	-	-	-	60	60	60		
10		20M CK(N) to TMOUTCK N	-	-	-	70	70	70		
10	tpd	20M CK(N+1) to TMOUTCK N+1	-	-	-	70	70	70	nS	12

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:37Z
Check the source to verify that this is the current version before use.

6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tsu	XNABLSRCN to 10M CK(N)	12	12	12	-	-	-	nS	12
		XDATSRCN to 10M CK(N)				-	-	-		
		XNABLSNKN to 10M CK(N)				-	-	-		
		XDATSNKN to 10M CK(N)				-	-	-		
		XNABLSRCN+1 to 10M CK(N+1)				-	-	-		
		XDATSRCN+1 to 10M CK(N+1)				-	-	-		
		XNABLSNKN+1 to 10M CK(N+1)				-	-	-		
11	tsu	XDATSNKN+1 to 10M CK(N+1)	12	12	12	-	-	-		
12	th	10M CK(N) to XNABLSRCN	10	10	10	-	-	-		
		10M CK(N) to XDATSRCN				-	-	-		
12	th	10M CK(N) to XNABLSNKN	10	10	10	-	-	-	nS	12

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:37Z
Check the source to verify that this is the current version before use.

6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
12	th	10M CK(N) to XDATSNKN	10	10	10	-	-	-	nS	12
		10M CK(N+1) to XNABLSRCN+1				-	-	-		
		10M CK(N+1) to XDATSRCN+1				-	-	-		
		10M CK(N+1) to XNABLSNKN+1				-	-	-		
12	th	10M CK(N+1) to XDATSNKN+1	10	10	10	-	-	-	nS	12

6.4 Module function. The MIL-STD-1397 type D serial interface module performs Manchester II encoding, decoding, and biasing. The module contains pulse transformers, delay lines, control drivers, control receiver and Manchester II encoder/decoder, receiver/transmitter circuits for channel N and N+1.

6.4.1 Serial North Atlantic Treaty Organization (NATO) Encoder/Decoder, Receiver/Transmitter (SNERT). The SNERT integrated circuit performs serial interface Manchester II encoding/decoding and transceiver functions. This circuit provides the two independent transceiver functions required (output and input), for one full-duplex, digital, serial Input/Output (I/O) channel. The encoder/transmitter portion of the SNERT chip accepts Transistor-Transistor Logic (TTL) clock timing, Non-Return-To-Zero (NRZ) data, and driver enable signals from the channel control logic, then generates the Manchester code signal for serial bit transmission at 10 megabits per second. The receiver/decoder portion of the chip receives the Manchester code signal from the I/O cable (via a signal transformer), decodes the serial data and clock information, then

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

outputs them as TTL compatible signals. This circuit also has the capability to loopback data and control signals from side A to side B and vice versa. This provides I/O channel test capability at the system level without disconnecting the I/O cables and reconnecting a jumper cable for test.

6.4.2 Delay line. The 10-75 nS delay line decodes the incoming Manchester encoded data. (The delay line is an analog delay terminated by 100 Ohms at 0.45 V.) The delay line driver signal (RCV OUTPUT) is an analog signal. The SNERT is designed to accept a 0.8 V amplitude signal with the dc level shifting with both temperature and supply voltages. These delay lines are physically situated close to the SNERT to minimize the channel crosstalk.

6.4.3 Description of I/O pins.

Signal name	Pin number	I/O	Description
20M CK(N+1)	8	I	20 MHZ CLK (CHAN N+1)
XNABLSRCN	11	I	TRANS ENBL SOURCE N
XNABLSNKN	12	I	TRANS ENBL SINK N
XDATSNKN	18	I	TRANS DATA SINK N
XNABLSRCN+1	28	I	TRANS ENBL SOURCE N+1
XDATSNKN+1	29	I	TRANS DATA SINK N+1
XNABLSNKN+1	62	I	TRANS ENBL SINK N+1
20M CK(N)	67	I	20 MHZ CLK (CHAN N)
LPBKMDEN+1	74	I	LOOPBACK TEST (CHAN N+1)
XDATSRCN+1	78	I	TRANS DATA SOURCE N+1
LPBKMDEN	80	I	LOOPBACK MODE N
XDATSRCN	99	I	TRANS DATA SOURCE N
PWR MACLR	58	I	PWR MA CLR

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:37Z
Check the source to verify that this is the current version before use.

6.4.3 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
RCVCKSNKN	9	O	RCV CLK SINK N
10M CK(N)	13	O	10 MHZ CLK (CHAN N)
RCVDTSNKN	15	O	RCV DATA SINK N
RCVENSNKN	17	O	RCV ENV SINK N
RCVDTSNKN+1	21	O	RCV DATA SINK N+1
RCVCKSNKN+1	26	O	RCV CLK SINK N+1
RCVENSNKN+1	30	O	RCV ENV SINK N+1
RCVCKSRCN	52	O	RCV CLK SOURCE N
RCVDTSRCN	61	O	RCV DATA SOURCE N
RCVENSRCN	63	O	RCV ENV SOURCE N
RCVDTSRCN+1	76	O	RCV DATA SOURCE N+1
RCVENSRCN+1	77	O	RCV ENV SOURCE N+1
TMOUTCK N	79	O	TIMEOUT CLK (CHAN N)
10M CK(N+1)	87	O	10 MHZ CLK (CHAN N+1)
TMOUTCK N+1	95	O	TIMEOUT CLK (CHAN N+1)
RCVCKSRCN+1	96	O	RCV CLK SOURCE N+1
RTN SNK N	42	O	RTN SINK N
SIG SNK N	43	O	SINK N SIGNAL
RTN SRC N	44	O	RTN SOURCE N
SIG SRC N	45	O	SOURCE N SIGNAL

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:37Z
Check the source to verify that this is the current version before use.

6.4.3 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
RTN SRC N+1	84	O	RTN SOURCE N+1
SIG SRC N+1	85	O	SOURCE N+1 SIGNAL
RTN SNK N+1	72	O	RTN SINK N+1
SIG SNK N+1	73	O	SINK N+1 SIGNAL
TYPE D N	41	I	TYPE D N
TYPE D N+1	37	I	TYPE D N+1

* 6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Surface Warfare Center (Code 603) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

* 6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:37Z
Check the source to verify that this is the currentversion before use.

Custodians:
Army - ER
Navy - NW
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

*Preparing activity:
Navy - NW

(Project 5963-0292)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C,2,3, 4,5,6, 7,14, 16,19, 22,23, 24,27, 32,33, 34,35, 36,38, 39,46, 47,48, 49,50, 53,54, 55,56, 57,59, 64,65, 66,68, 69,70, 71,82, 83,86, 88,89, 91,92, 93,94, 97,98, 100	Frame	10	INF	MOhm	1-9

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A2	Isolation, used to unused pins	C	2,3,4, 5,6,7, 14,16, 19,22, 23,24, 27,32, 33,34, 35,36, 38,39, 46,47, 48,49, 50,53, 54,55, 56,57, 59,64, 65,66, 68,69, 70,71, 82,83, 86,88, 89,91, 92,93, 94,97, 98,100	10	INF	MOhm	1-9
A3-A51	Isolation, unused to unused pins	See note 4		10	INF	MOhm	1-9

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1-A51
Isolation measurement	NA	± 500 KOhm	A1-A51

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. The unused pins on this module are: 2,3,4,5,6,7,14,16,19, 22,23,24,27,32,33,34,35,36,38,39,46,47,48,49,50,53,54,55,56,57, 59,64,65,66,68,69,70,71,82,83,86,88,89,91,92,93,94,97,98 and 100. When testing for unused to unused pin isolation, connect one unused pin to A and the rest of the unused pins to B.

5. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

6. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

7. Caution: Set megohm bridge to limit current to 10 μA before applying voltage to the module.

8. The test circuit shall be as indicated on figure 6.

9. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

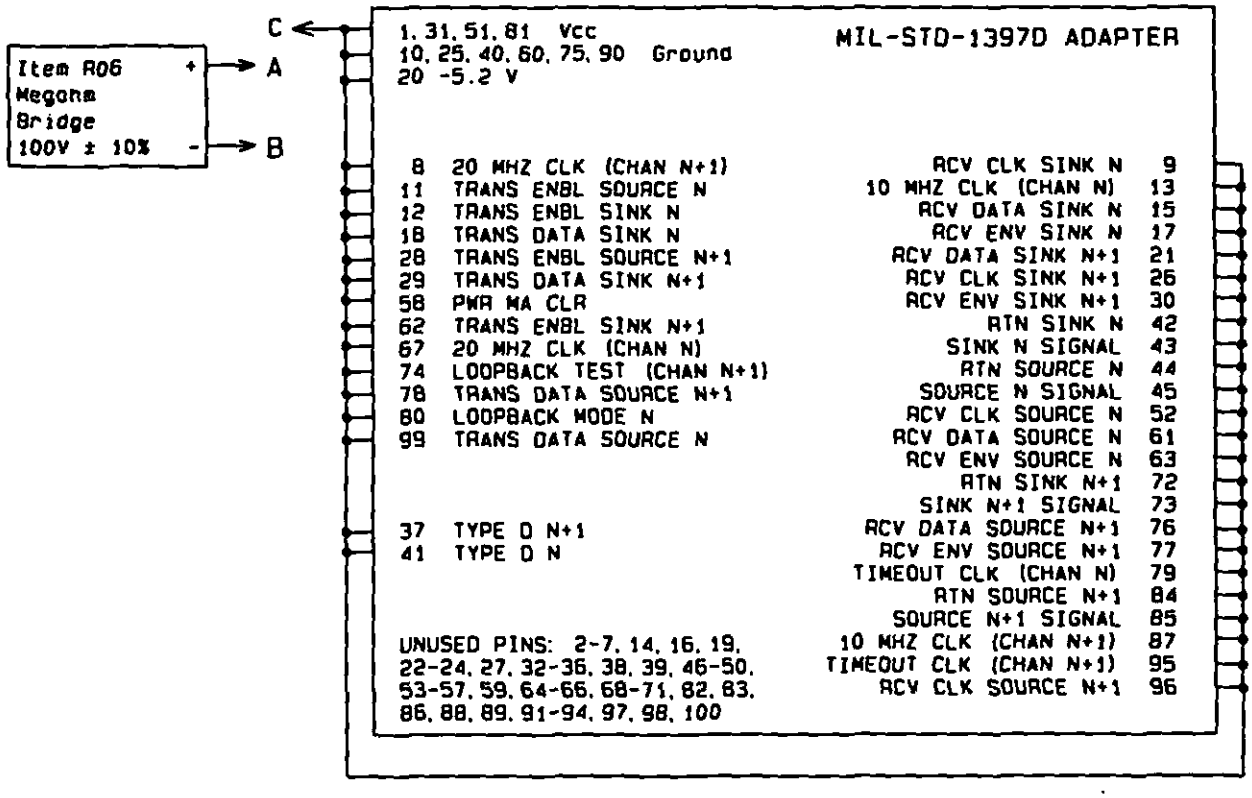


FIGURE 6. Isolation test circuit for table II.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:37Z
Check the source to verify that this is the current version before use.

* TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, Vcc	1	31				
B6	Continuity, Vcc	51	81				
B7	Continuity, TYPE D N+1 to TYPE D N	37	41	0	1	Ohm	1-5

NOTES:

*1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B7

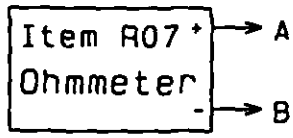
2. The ohmmeter used shall be item R07.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:37Z
Check the source to verify that this is the current version before use.



1, 31, 51, 81 Vcc
10, 25, 40, 60, 75, 90 Ground
20 -5.2 V

MIL-STD-1397D ADAPTER

8	20 MHZ CLK (CHAN N+1)	RCV CLK SINK N	9
11	TRANS ENBL SOURCE N	10 MHZ CLK (CHAN N)	13
12	TRANS ENBL SINK N	RCV DATA SINK N	15
18	TRANS DATA SINK N	RCV ENV SINK N	17
28	TRANS ENBL SOURCE N+1	RCV DATA SINK N+1	21
29	TRANS DATA SINK N+1	RCV CLK SINK N+1	26
58	PWR MA CLR	RCV ENV SINK N+1	30
62	TRANS ENBL SINK N+1	RTN SINK N	42
67	20 MHZ CLK (CHAN N)	SINK N SIGNAL	43
74	LOOPBACK TEST (CHAN N+1)	RTN SOURCE N	44
78	TRANS DATA SOURCE N+1	SOURCE N SIGNAL	45
80	LOOPBACK MODE N	RCV CLK SOURCE N	52
99	TRANS DATA SOURCE N	RCV DATA SOURCE N	61
		RCV ENV SOURCE N	63
		RTN SINK N+1	72
		SINK N+1 SIGNAL	73
37	TYPE D N+1	RCV DATA SOURCE N+1	76
41	TYPE D N	RCV ENV SOURCE N+1	77
		TIMEOUT CLK (CHAN N)	79
		RTN SOURCE N+1	84
		SOURCE N+1 SIGNAL	85
UNUSED PINS: 2-7, 14, 16, 19, 22-24, 27, 32-36, 38, 39, 46-50, 53-57, 59, 64-66, 68-71, 82, 83, 86, 88, 89, 91-94, 97, 98, 100		10 MHZ CLK (CHAN N+1)	87
		TIMEOUT CLK (CHAN N+1)	95
		RCV CLK SOURCE N+1	96

FIGURE 7. Continuity test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:37Z
Check the source to verify that this is the current version before use.

TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6
* C	V2	7.0	20	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

*1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
(V2)	-7.0 V	+70 mV,-0 mV	C
Power supply current limit, (Vcc)	1.91 A	NA	C
(V2)	-0.47 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

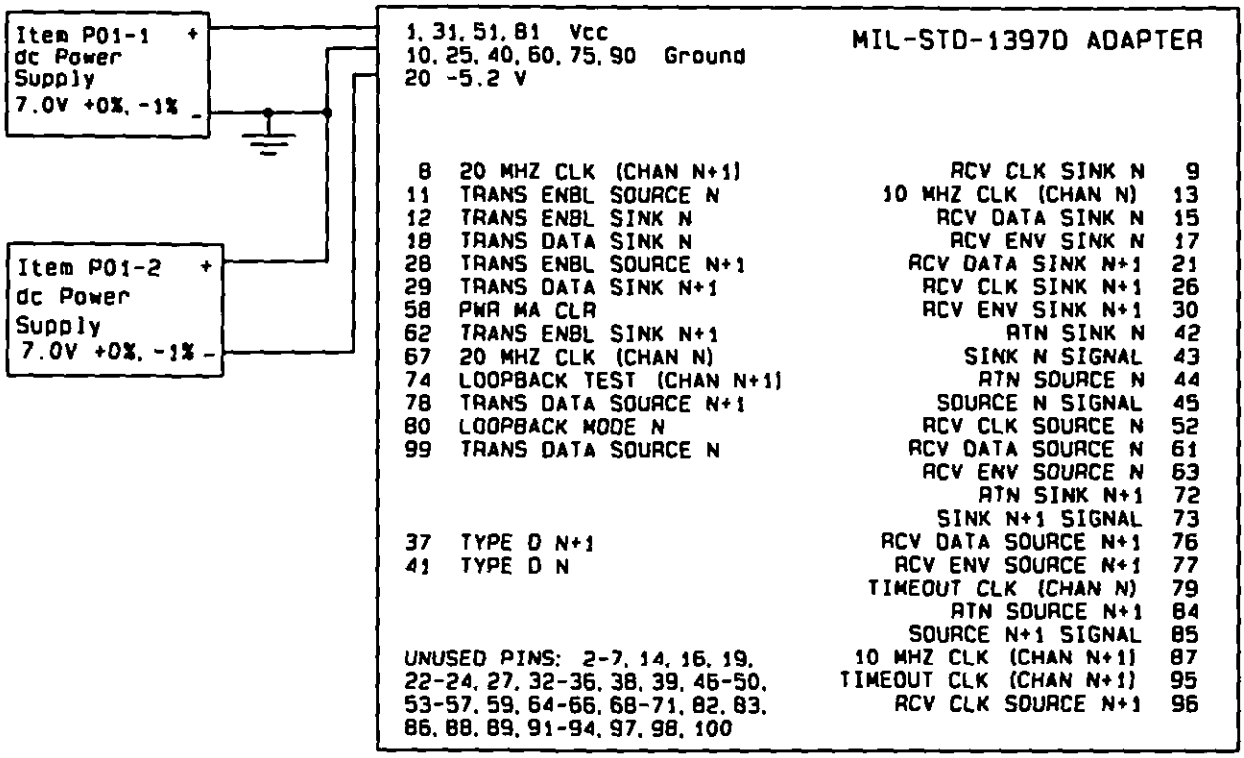
3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:37Z
Check the source to verify that this is the current version before use.



* FIGURE 8. Power supply stress test circuit for table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test.

	Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
	D1	20M CK(N+1)	8	5.5	-0.022	1.02	mA	1-3
	D2	XNABLSRCN	11		0.0	0.1		
	D3	XNABLSNKN	12					
	D4	XDATSNKN	18					
	D5	XNABLSRCN+1	28					
	D6	XDATSNKN+1	29		0.0	0.1		
*	D7	PWR MACLR	58		-0.582	1.14		
*	D8	XNABLSNKN+1	62		0.0	0.1		
*	D9	20M CK(N)	67		-0.022	1.02		
*	D10	LPBKMDEN+1	74		-0.022	1.02		
*	D11	XDATSRCN+1	78		0.0	0.1		
*	D12	LPBKMDEN	80		-0.022	1.02		
*	D13	XDATSRCN	99	5.5	0.0	0.1	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:37Z
Check the source to verify that this is the current version before use.

NOTES:

*1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D13
(V2)	-5.7 V	± 57 mV	D1-D13
Power supply current limit, (Vcc)	1.59 A	NA	D1-D13
(V2)	-0.39 A	NA	D1-D13
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D13
Pattern voltage high, (VH)	3.0 V	± 100 mV	D1-D13
low, (VL)	0.0 V	± 100 mV	D1-D13
Input current measurement	NA	± 52 µA	D1
		± 5 µA	D2-D6
		± 86 µA	D7
		± 5 µA	D8
		± 52 µA	D9,D10
		± 5 µA	D11
		± 52 µA	D12
	NA	± 5 µA	D13

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:37Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	RCVCKSNKN	9	0.0	-100.0	-40.0	mA	1-4
E2	10M CK(N)	13					
E3	RCVDTSNKN	15					
E4	RCVENSNKN	17					
E5	RCVDTSNKN+1	21					
E6	RCVCKSNKN+1	26					
E7	RCVENSNKN+1	30					
E8	RCVCKSRCN	52					
E9	RCVDTSRCN	61					
E10	RCVENSRCN	63					
E11	RCVDTSRCN+1	76					
E12	RCVENSRCN+1	77					
E13	TMOUTCK N	79					
E14	10M CK(N+1)	87					
E15	TMOUTCK N+1	95					
E16	RCVCKSRCN+1	96	0.0	-100.0	-40.0	mA	1-4

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:37Z
Check the source to verify that this is the current version before use.

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V2)	5.5 V -5.7 V	± 55 mV ± 57 mV	E1-E16 E1-E16
Power supply current limit, (Vcc) (V2)	1.59 A -0.39 A	NA NA	E1-E16 E1-E16
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E16
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	E1-E16 E1-E16
Output current measurement	NA	± 3 mA	E1-E16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:37Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

TABLE VII. Input current, high level, test.

	Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
	F1	20M CK(N+1)	8	2.4	-3.04	-0.503	mA	1-3
	F2	XNABLSRCN	11		-0.4	0.02		
	F3	XNABLSNKN	12					
	F4	XDATSNKN	18					
	F5	XNABLSRCN+1	28					
	F6	XDATSNKN+1	29		-0.4	0.02		
*	F7	PWR MACLR	58		-6.64	-0.353		
	F8	XNABLSNKN+1	62		-0.4	0.02		
	F9	20M CK(N)	67		-3.04	-0.503		
	F10	LPBKMDEN+1	74		-3.04	-0.503		
	F11	XDATSRCN+1	78		-0.4	0.02		
	F12	LPBKMDEN	80		-3.04	-0.503		
	F13	XDATSRCN	99	2.4	-0.4	0.02	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:37Z
Check the source to verify that this is the current version before use.

NOTES:

*1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F13
(V2)	-5.7 V	± 57 mV	F1-F13
Power supply current limit, (Vcc)	1.59 A	NA	F1-F13
(V2)	-0.39 A	NA	F1-F13
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F13
Pattern voltage high, (VH)	3.0 V	± 100 mV	F1-F13
low, (VL)	0.0 V	± 100 mV	F1-F13
Input current measurement	NA	± 127 µA	F1
		± 21 µA	F2-F6
		± 314 µA	F7
		± 21 µA	F8
		± 127 µA	F9,F10
		± 21 µA	F11
		± 127 µA	F12
	NA	± 21 µA	F13

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	20M CK(N+1)	8	0.4	-3.04	-0.503	mA	1-3
G2	XNABLSRCN	11		-0.4	0.02		
G3	XNABLSNKN	12					
G4	XDATSNKN	18					
G5	XNABLSRCN+1	28					
G6	XDATSNKN+1	29		-0.4	0.02		
* G7	PWR MACLR	58		-6.64	-0.353		
G8	XNABLSNKN+1	62		-0.4	0.02		
G9	20M CK(N)	67		-3.04	-0.503		
G10	LPBKMDEN+1	74		-3.04	-0.503		
G11	XDATSRCN+1	78		-0.4	0.02		
G12	LPBKMDEN	80		-3.04	-0.503		
G13	XDATSRCN	99	0.4	-0.4	0.02	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

NOTES:

*1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G13
(V2)	-5.7 V	± 57 mV	G1-G13
Power supply current limit, (Vcc)	1.59 A	NA	G1-G13
(V2)	-0.39 A	NA	G1-G13
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G13
Pattern voltage high, (VH)	3.0 V	± 100 mV	G1-G13
low, (VL)	0.0 V	± 100 mV	G1-G13
Input current measurement	NA	± 127 µA	G1
		± 21 µA	G2-G6
		± 314 µA	G7
		± 21 µA	G8
		± 127 µA	G9,G10
		± 21 µA	G11
		± 127 µA	G12
	NA	± 21 µA	G13

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

Source: https://assist.dla.mil -- Downloaded: 20261-20T18:37Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	RCVCKSNKN	9	-0.4	2.4	4.5	V	1-4
H2	10M CK(N)	13					
H3	RCVDTSNKN	15					
H4	RCVENSNKN	17					
H5	RCVDTSNKN+1	21					
H6	RCVCKSNKN+1	26					
H7	RCVENSNKN+1	30					
H8	RCVCKSRCN	52					
H9	RCVDTSRCN	61					
H10	RCVENSRCN	63					
H11	RCVDTSRCN+1	76					
H12	RCVENSRCN+1	77					
H13	TMOUTCK N	79					
H14	10M CK(N+1)	87					
H15	TMOUTCK N+1	95					
H16	RCVCKSRCN+1	96	-0.4	2.4	4.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:37Z
Check the source to verify that this is the current version before use.

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V2)	4.5 V -4.7 V	± 55 mV ± 47 mV	H1-H16 H1-H16
Power supply current limit, (Vcc) (V2)	1.59 A -0.39 A	NA NA	H1-H16 H1-H16
Forcing current, (Ioh)	-0.4 mA	± 40 µA	H1-H16
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H16 H1-H16
Output voltage measurement	NA	± 105 mV	H1-H16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:37Z
Check the source to verify that this is the currentversion before use.

NOTES: - Continued.

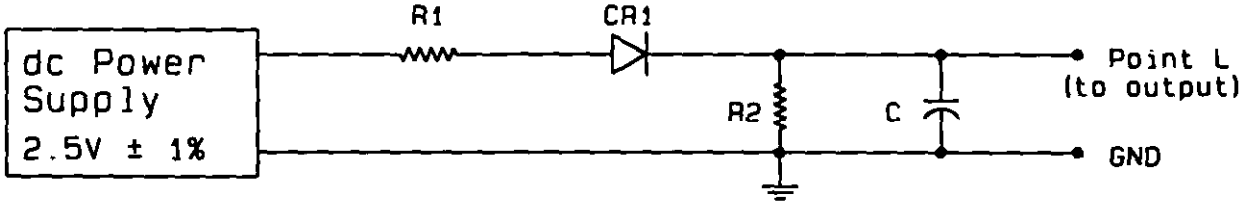
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.



NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

    (a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

    (b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

    (c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

    (d) C not required for any other tests.

FIGURE 9. Logic load circuit.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:37Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are specified below.

4. The value of R2 for each output load shall be as specified below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
9,13,15,17,21, 26,30,52,61,63, 76,77,79,87,95, 96	6200	6.0	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	RCVCKSNKN	9	6.0	0.0	0.5	V	1-4
J2	10M CK(N)	13					
J3	RCVDTSNKN	15					
J4	RCVENSNKN	17					
J5	RCVDTSNKN+1	21					
J6	RCVCKSNKN+1	26					
J7	RCVENSNKN+1	30					
J8	RCVCKSRCN	52					
J9	RCVDTSRCN	61					
J10	RCVENSRCN	63					
J11	RCVDTSRCN+1	76					
J12	RCVENSRCN+1	77					
J13	TMOUTCK N	79					
J14	10M CK(N+1)	87					
J15	TMOUTCK N+1	95					
J16	RCVCKSRCN+1	96	6.0	0.0	0.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:37Z
Check the source to verify that this is the current version before use.

NOTES:

*1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V2)	4.5 V -4.7 V	± 55 mV ± 47 mV	J1-J16 J1-J16
Power supply current limit, (Vcc) (V2)	1.59 A -0.39 A	NA NA	J1-J16 J1-J16
Forcing current, (Iol)	6.0 mA	± 600 µA	J1-J16
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J16 J1-J16
Output voltage measurement	NA	± 25 mV	J1-J16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:37Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:37Z
Check the source to verify that this is the current version before use.

TABLE XI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	XNABLSRCN	11	-18.0	-1.5	0.0	V	1-3
L2	XNABLSNKN	12					
L3	XDATSNKN	18					
L4	XNABLSRCN+1	28					
L5	XDATSNKN+1	29					
L6	XNABLSNKN+1	62					
L7	XDATSRCN+1	78					
L8	XDATSRCN	99	-18.0	-1.5	0.0	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L8
(V2)	Off	NA	L1-L8
Power supply current limit, (Vcc)	NA	NA	L1-L8
(V2)	NA	NA	L1-L8
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L8

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:37Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L8 L1-L8
Input voltage measurement	NA	± 75 mV	L1-L8

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

TABLE XII. Power supply filter capacitance test.

	Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
*	M1	PSFC	1	1.9	3.6	µF	1-4
*	M2	PSFC	51	3.8	7.2	µF	1-4
*	M3	PSFC	20	14.4	26.7	µF	1-4

NOTES:

*1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1-M3
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1-M3
Capacitance measurement	NA | NA	± 85 nF ± 170 nF ± 615 nF	M1 M2 M3

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

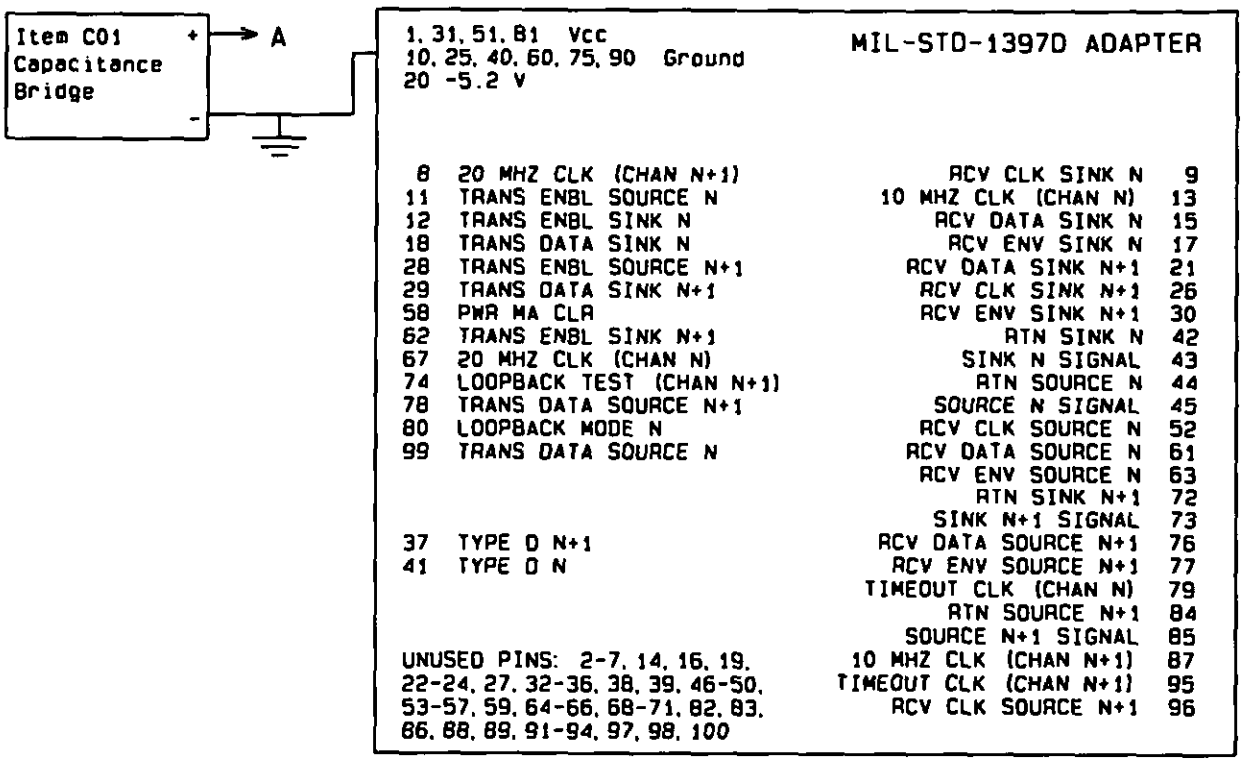


FIGURE 10. Power supply filter capacitance test circuit for table XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

TABLE XIII. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
* N1	Icc	1,31, 51,81	450	1150	mA	1-3
N2	Icc	20	-263	-130	mA	1-3

NOTES:

*1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N1,N2
(V2)	-5.7 V	± 57 mV	N1,N2
Power supply current limit, (Vcc)	1.59 A	NA	N1,N2
(V2)	-0.39 A	NA	N1,N2
Pattern voltage high, (VH)	3.0 V	± 100 mV	N1,N2
low, (VL)	0.0 V	± 100 mV	N1,N2
Power supply current measurement	NA	± 35 mA	N1
	NA	± 6.65 mA	N2

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIII.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:37Z
Check the source to verify that this is the currentversion before use.

TABLE XIV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (100 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (100 nS)	1-3

NOTES:

*1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	P1
(V2)	-4.7 V	± 47 mV	P1
(Vcc)	5.5 V	± 55 mV	P2
(V2)	-5.7 V	± 57 mV	P2
Power supply current limit, (Vcc)	1.59 A	NA	P1,P2
(V2)	-0.39 A	NA	P1,P2
Pattern voltage high, (VH)	2.4 V	± 100 mV	P1,P2
low, (VL)	0.4 V	± 100 mV	P1,P2
Output voltage verification high, (Voh)	NA	± 105 mV	P1
high, (Voh)	|	± 155 mV	P2
low, (Vol)	NA	± 20 mV	P1,P2

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:37Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

* (b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XIV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

For pins 42,43,44,45,72,73,84,85:

High level: 1.0 V ≤ Voh ≤ 3.25 V
Low level: -3.25 V ≤ Vol ≤ -1.0 V

For pins 9,13,15,17,21,26,30,52,61,63,76,77,79,87,95,96:

High level: 2.4 V ≤ Voh ≤ Vcc
Low level: 0.0 V ≤ Vol ≤ 0.5 V

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:37Z
Check the source to verify that this is the current version before use.

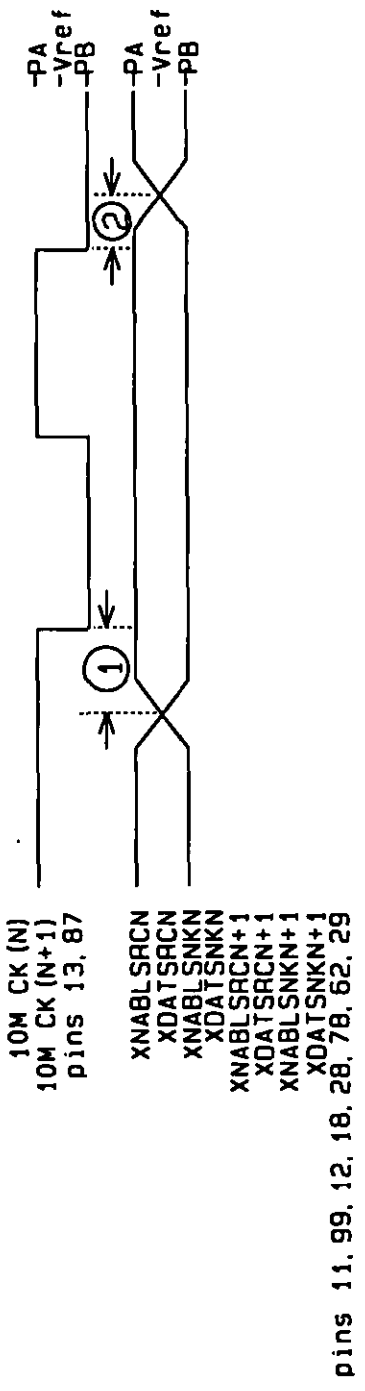


FIGURE 11. Input timing parameters for table XIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

NOTES:

1. (1) XNABLSRCN to 10M CK(N). Setup time: 12 nS minmum.
2. (1) XDATSRCN to 10M CK(N). Setup time: 12 nS minimum.
3. (1) XNABLSNKN to 10M CK(N). Setup time: 12 nS minimum.
4. (1) XDATSNKN to 10M CK(N). Setup time: 12 nS minimum.
5. (1) XNABLSRCN+1 to 10M CK(N+1). Setup time: 12 nS minimum.
6. (1) XDATSRCN+1 to 10M CK(N+1). Setup time: 12 nS minimum.
7. (1) XNABLSNKN+1 to 10M CK(N+1). Setup time: 12 nS minimum.
8. (1) XDATSNKN+1 to 10M CK(N+1). Setup time: 12 nS minimum.
9. (2) 10M CK(N) to XNABLSRCN. Hold time: 10 nS minimum.
10. (2) 10M CK(N) to XDATSRCN. Hold time: 10 nS minimum.
11. (2) 10M CK(N) to XNABLSNKN. Hold time: 10 nS minimum.
12. (2) 10M CK(N) to XDATSNKN. Hold time: 10 nS minimum.
13. (2) 10M CK(N+1) to XNABLSRCN+1. Hold time: 10 nS minimum.
14. (2) 10M CK(N+1) to XDATSRCN+1. Hold time: 10 nS minimum.
15. (2) 10M CK(N+1) to XNABLSNKN+1. Hold time: 10 nS minimum.
16. (2) 10M CK(N+1) to XDATSNKN+1. Hold time: 10 nS minimum.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T18:37Z
Check the source to verify that this is the currentversion before use.

TABLE XV. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (100 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (100 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (100 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (100 nS) maximum power supply	1-6

NOTES:

*1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	Q1,Q3
(V2)	-4.7 V	± 47 mV	Q1,Q3
(Vcc)	5.5 V	± 55 mV	Q2,Q4
(V2)	-5.7 V	± 57 mV	Q2,Q4
Power supply current limit, (Vcc)	1.59 A	NA	Q1-Q4
(V2)	-0.39 A	NA	Q1-Q4

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:37Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input voltage			
high, (VH)	2.4 V	± 100 mV	Q1-Q4
low, (VL)	0.4 V	± 100 mV	Q1-Q4
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	|	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

* (b) The output compare levels shall be as follows:

For pins 42,43,44,45,72,73,84,85:

High level: 1.0 V ≤ Voh ≤ 3.25 V
Low level: -3.25 V ≤ Vol ≤ -1.0 V

For pins 9,13,15,17,21,26,30,52,61,63,76,77,79,87,95,96:

High level: Vref ≤ Voh ≤ Vcc
Low level: 0.0 V ≤ Vol ≤ Vref

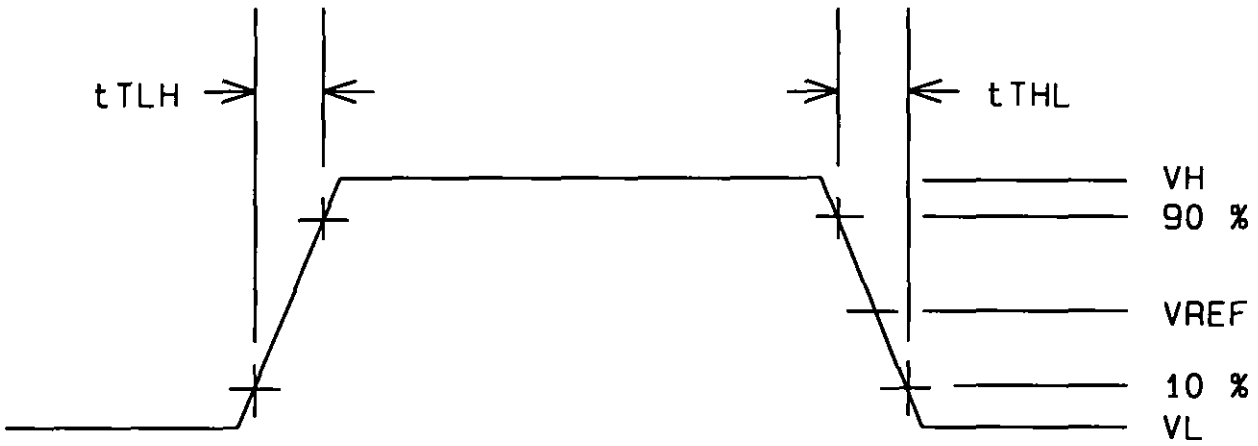
(c) The dynamic parameters are found in table XVI.

3. The load circuit for this test is specified on figure 9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

* 4. I/O waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



Where:

(a) VH and VL are as specified in note 2(b).

(b) tTLH = tTHL ≤ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XV.

6. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

TABLE XVI. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	RTN SRC N to RCVENSRCN	-	-	-	56	56	56	nS	12
		SIG SRC N to RCVENSRCN	-	-	-					
		RTN SRC N+1 to RCVENSRCN+1	-	-	-					
1		SIG SRC N+1 to RCVENSRCN+1	-	-	-	56	56	56		
2		RTN SRC N to RCVENSRCN	-	-	-	206	206	206		
		SIG SRC N to RCVENSRCN	-	-	-					
		RTN SRC N+1 to RCVENSRCN+1	-	-	-					
2		SIG SRC N+1 to RCVENSRCN+1	-	-	-	206	206	206		
3		RTN SRC N to RCVCKSRCN	-	-	-	176	176	176		
3	tpd	SIG SRC N to RCVCKSRCN	-	-	-	176	176	176	nS	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	RTN SRC N+1 to RCVCKSRCN+1	-	-	-	176	176	176	nS	12
3		SIG SRC N+1 to RCVCKSRCN+1	-	-	-	176	176	176		
4		RTN SNK N to RCVENSNKN	-	-	-	56	56	56		
		SIG SNK N to RCVENSNKN	-	-	-					
		RTN SNK N+1 to RCVENSNKN+1	-	-	-					
4		SIG SNK N+1 to RCVENSNKN+1	-	-	-	56	56	56		
5		RTN SNK N to RCVENSNKN	-	-	-	206	206	206		
		SIG SNK N to RCVENSNKN	-	-	-					
		RTN SNK N+1 to RCVENSNKN+1	-	-	-					
5	tpd	SIG SNK N+1 to RCVENSNKN+1	-	-	-	206	206	206	nS	12

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:37Z
Check the source to verify that this is the current version before use.

TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	RTN SNK N to RCVCKSNKN	-	-	-	176	176	176	nS	12
		SIG SNK N to RCVCKSNKN	-	-	-					
		RTN SNK N+1 to RCVCKSNKN+1	-	-	-					
6		SIG SNK N+1 to RCVCKSNKN+1	-	-	-	176	176	176		
7		20M CK(N) to 10M CK(N)	-	-	-	50	50	50		
7		20M CK(N+1) to 10M CK(N+1)	-	-	-	50	50	50		
8		20M CK(N) to 10M CK(N)	-	-	-	59	59	59		
8		20M CK(N+1) to 10M CK(N+1)	-	-	-	59	59	59		
9		20M CK(N) to TMOUTCK N	-	-	-	60	60	60		
9		20M CK(N+1) to TMOUTCK N+1	-	-	-	60	60	60		
10	tpd	20M CK(N) to TMOUTCK N	-	-	-	70	70	70	nS	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
10	tpd	20M CK(N+1) to TMOUTCK N+1	-	-	-	70	70	70	nS	12
11	tsu	XNABLSRCN to 10M CK(N)	12	12	12	-	-	-		
		XDATSRCN to 10M CK(N)				-	-	-		
		XNABLSNKN to 10M CK(N)				-	-	-		
		XDATSNKN to 10M CK(N)				-	-	-		
		XNABLSRCN+1 to 10M CK(N+1)				-	-	-		
		XDATSRCN+1 to 10M CK(N+1)				-	-	-		
		XNABLSNKN+1 to 10M CK(N+1)				-	-	-		
11	tsu	XDATSNKN+1 to 10M CK(N+1)	12	12	12	-	-	-		
12	th	10M CK(N) to XNABLSRCN	10	10	10	-	-	-		
12	th	10M CK(N) to XDATSRCN	10	10	10	-	-	-	nS	12

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:37Z
Check the source to verify that this is the current version before use.

TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
12	th	10M CK(N) to XNABLSNKN	10	10	10	-	-	-	nS	12
		10M CK(N) to XDATSNKN				-	-	-		
		10M CK(N+1) to XNABLSRCN+1				-	-	-		
		10M CK(N+1) to XDATSRCN+1				-	-	-		
		10M CK(N+1) to XNABLSNKN+1				-	-	-		
12	th	10M CK(N+1) to XDATSNKN+1	10	10	10	-	-	-	nS	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

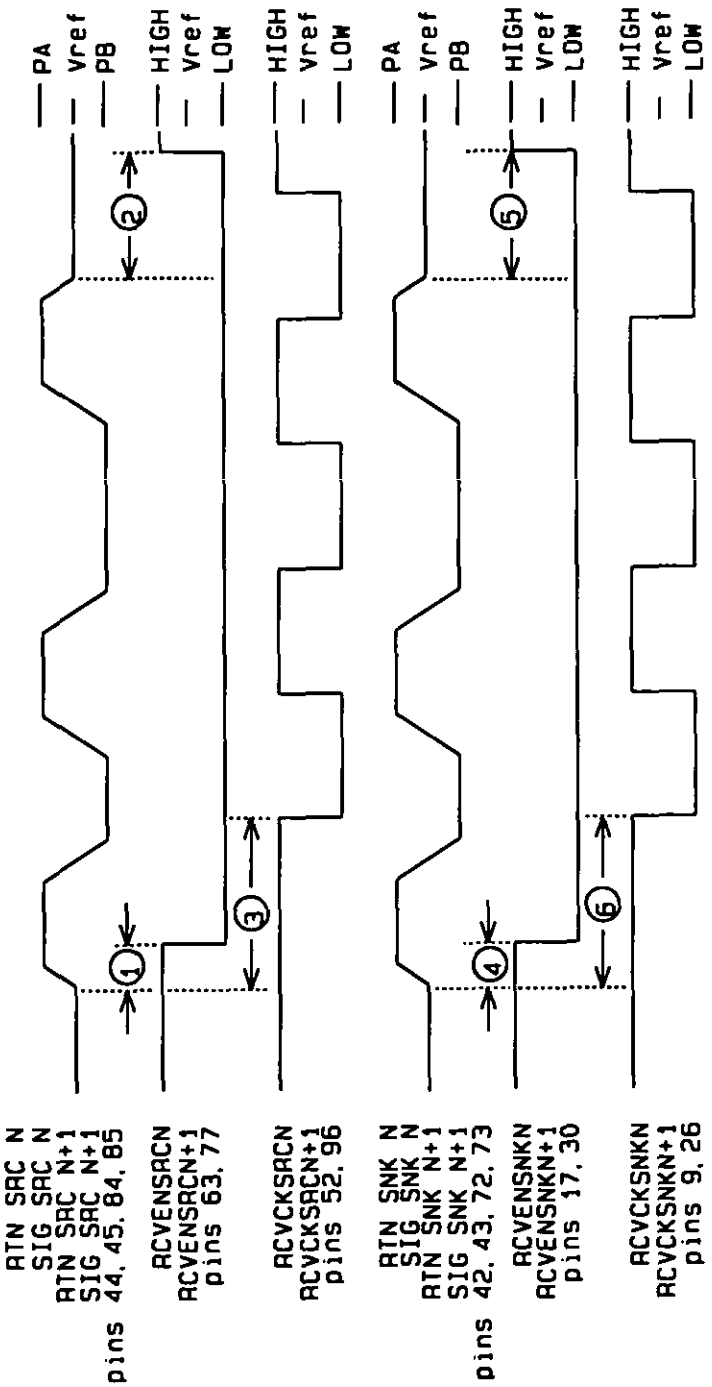


FIGURE 12a. Dynamic waveforms for table XVI.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:37Z
Check the source to verify that this is the current version before use.

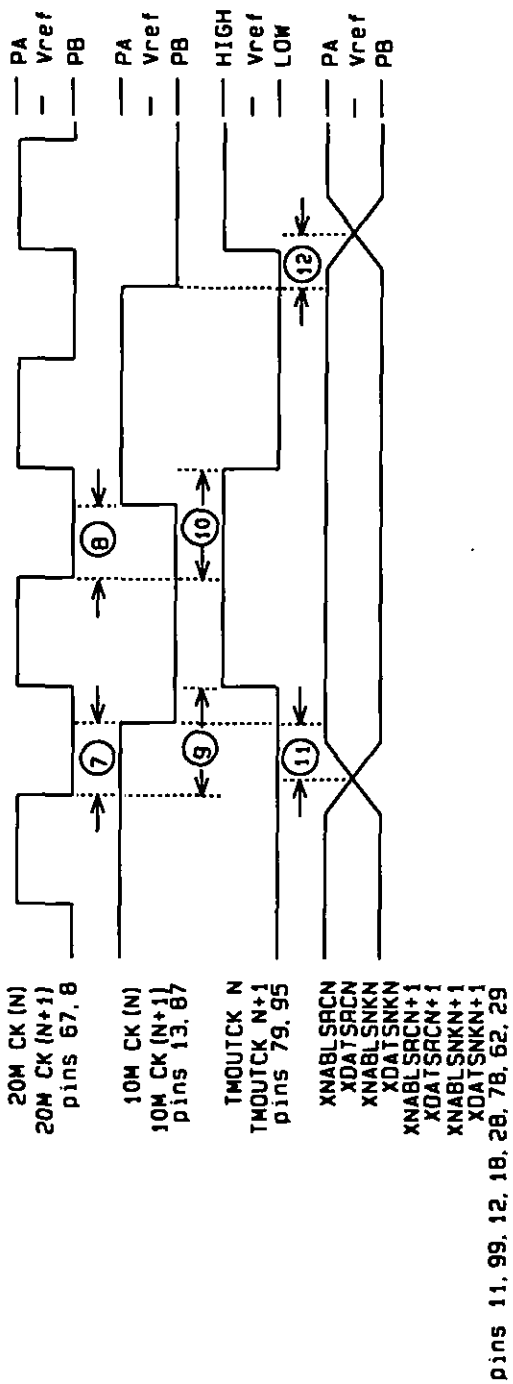


FIGURE 12b. Dynamic waveforms for table XVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

TABLE XVII. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
R1	20M CK(N+1)	8	2	75	pF	1-4
R2	XNABLSRCN	11				
R3	XNABLSNKN	12				
R4	XDATSNKN	18				
R5	XNABLSRCN+1	28				
R6	XDATSNKN+1	29	2	75		
R7	PWR MACLR	58	8	200		
R8	XNABLSNKN+1	62	2	100		
R9	20M CK(N)	67	4	75		
R10	LPBKMDEN+1	74	4	100		
R11	XDATSRCN+1	78	2	75		
R12	LPBKMDEN	80	4	100		
R13	XDATSRCN	99	2	125	pF	1-4

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:37Z
Check the source to verify that this is the current version before use.

NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R13
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R13
Capacitance measurement	NA NA	± 3.65 pF ± 9.6 pF ± 3.65 pF ± 3.55 pF ± 4.8 pF ± 3.65 pF ± 4.8 pF ± 6.15 pF	R1-R6 R7 R8 R9 R10 R11 R12 R13

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:37Z
Check the source to verify that this is the current version before use.

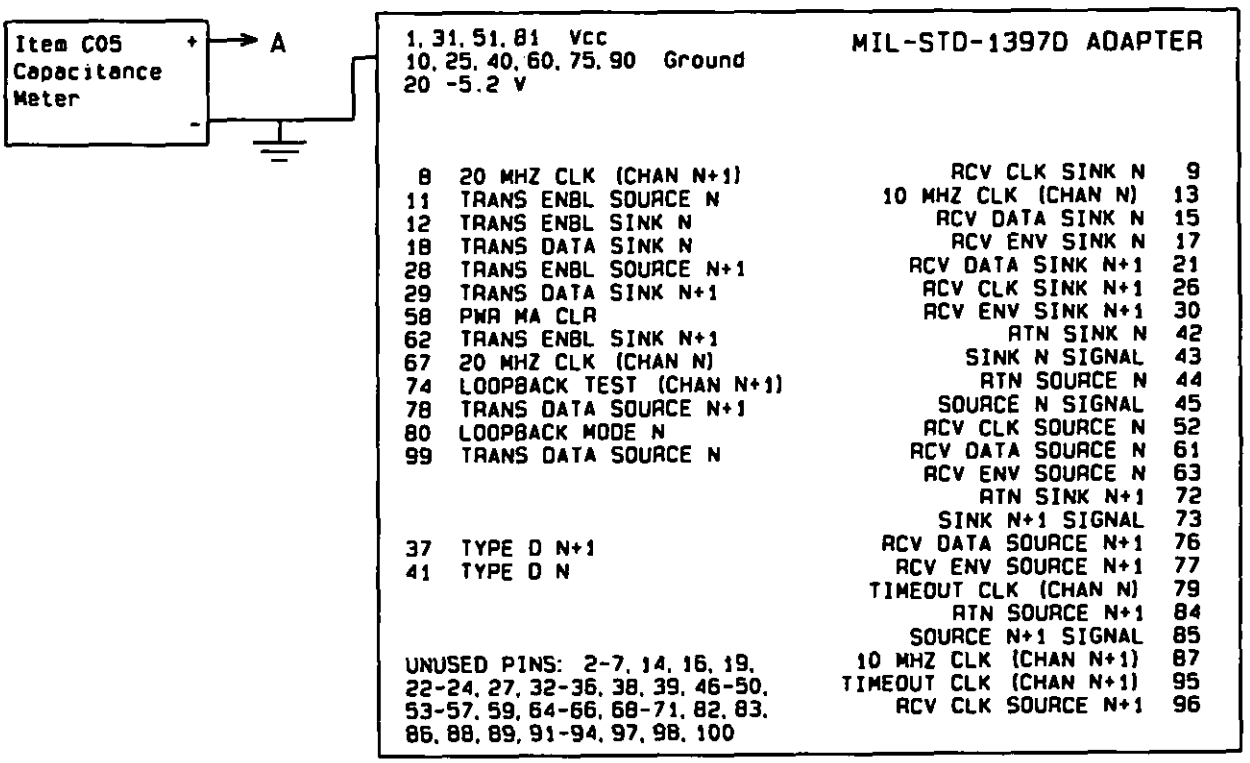


FIGURE 13. Input capacitance test circuit for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA	1	0	400	mV	1-4
T2	PSTA	20	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V2)	5.0 V -5.2 V	± 250 mV ± 260 mV	T1,T2 T1,T2
Power supply current limit, (Vcc) (V2)	1.59 A -0.39 A	NA NA	T1,T2 T1,T2
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:37Z
Check the source to verify that this is the currentversion before use.

NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XVIII.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XVIII.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

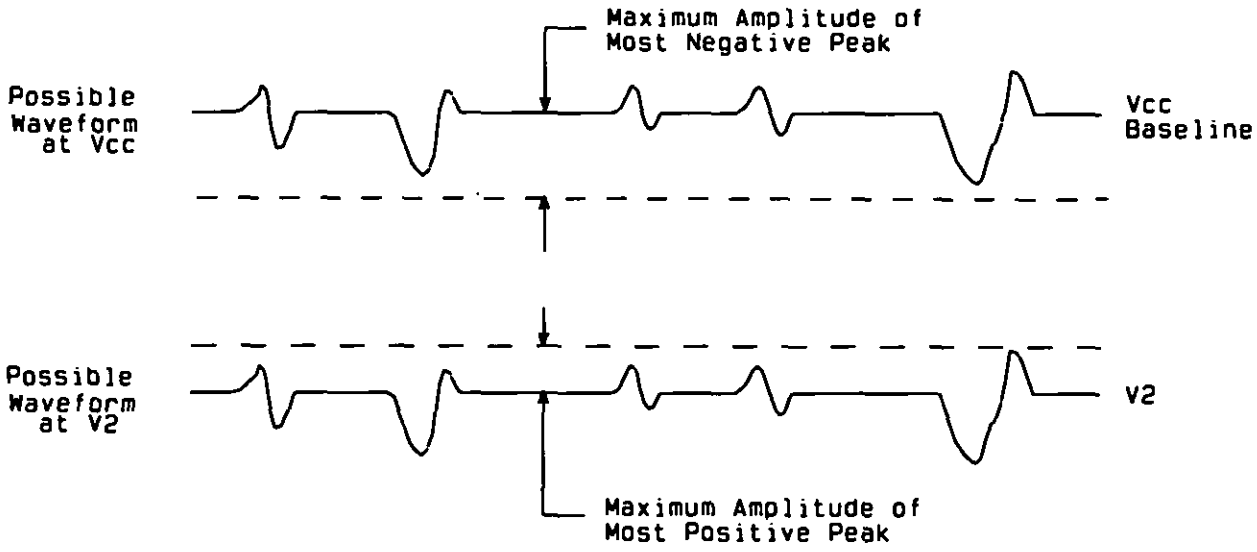
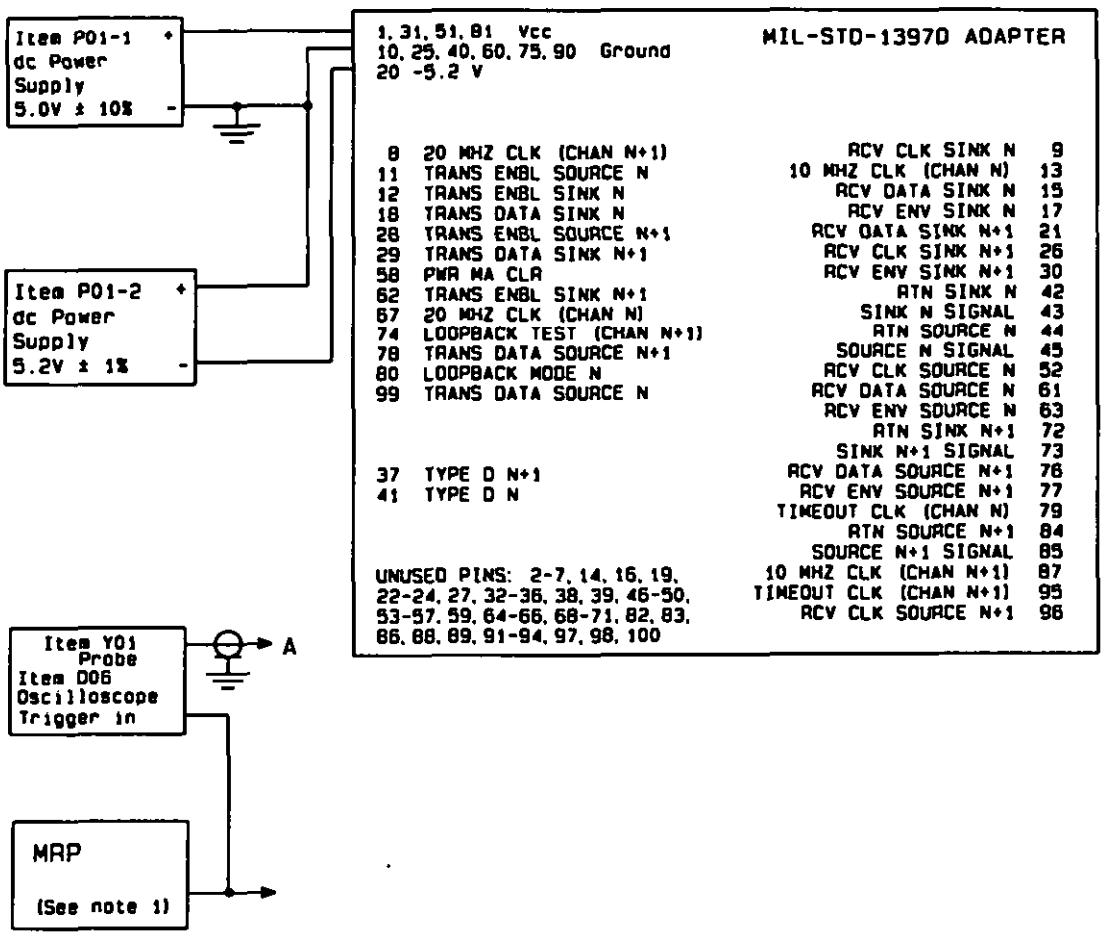


FIGURE 14. Power supply transient amplitude waveform for table XVIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.



NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XVIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:37Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-M-28787/529A	2. DOCUMENT DATE (YYMMDD) 1994 March 23

3. DOCUMENT TITLE: Modules, Standard Electronic, MIL-STD-1397 Type D Serial Interface Adapter Module, Key Code AJL

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Point of Contact (TPOC): Mr. Joseph P. Toy (Code 603A) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial TPOC: 812-854-5993	(2) AUTOVON 88-482-5993
c. ADDRESS (Include Zip Code) Commander ATTN: Joseph P. Toy, Code 603A, Bldg. 2917 NAVSURFWARCENDIV, 300 Highway 361 Crane, IN 47522-5001	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil - Downloaded: 2020640-20T18:37Z
Check the source to verify that this is the current version before use.